WildfireWatch: Project Guide & Full Video Script

Focus: Live Satellite/Weather APIs + Machine Learning & Physics Forecasting Pipeline

1. Main Live APIs We Use

- **1. NASA FIRMS Area REST API (Live Satellite Thermal Data):** Fetches real-time VIIRS S-NPP (375m resolution) and MODIS (1km resolution) thermal anomalies globally. Provides Latitude, Longitude, Fire Radiative Power (FRP in MW), Brightness Temperature (Kelvin), and UTC Acquisition Time.

- **2. Element84 Earth Search STAC API (Sentinel-2 L2A Satellite Scenes):** Queries the latest Sentinel-2 L2A satellite imagery pass over the active wildfire bounding box to feed Short-Wave Infrared (SWIR) verification.

- **3. Open-Meteo Weather Forecast API (Live Meteorology):** Fetches real-time hourly 10m Wind Speed (km/h), Wind Direction (degrees), 2m Relative Humidity (%), and Air Temperature (C) at the exact fire centroid.

- **4. Open-Meteo Digital Elevation Model (DEM) API (Live Terrain Slope):** Samples 5 elevation points (Center, North, South, East, West at 1 km offset) to calculate the terrain slope angle in degrees and uphill aspect.

2. Machine Learning & Physics Models Used

- **1. DBSCAN Spatial Clustering (Unsupervised ML):** Raw satellite data gives scattered heat dots. We run Density-Based Spatial Clustering (DBSCAN) with a 15 km Haversine radius to automatically group individual thermal anomalies into discrete Wildfire Incidents and extract the T+0h active fire polygon.

- **2. Sentinel-2 U-Net Transfer Learning & Active Fire Index (Deep Learning / Computer Vision):** Uses a U-Net convolutional encoder-decoder architecture trained via transfer learning from Landsat-8 to Sentinel-2 Short-Wave Infrared channels (B12 2190nm, B11 1610nm, B8A 865nm). Since SWIR light penetrates smoke clouds, this step separates true active burning fronts from smoke plumes and false alarms.

- **3. 100-Tree Random Forest Spread Ensemble (Supervised ML Regression):** Takes 6 live features - Wind Speed, Relative Humidity, Air Temperature, Terrain Slope, Fire Radiative Power (FRP), and SWIR Active Fire Index - across 100 decision trees to predict the Rate of Spread (in meters/minute) and calculate the Forecast Confidence Score (%) from tree agreement.

- **4. Rothermel Elliptical Wavelet Propagation + Hindcast Validation:** Projects the Random Forest spread rate outward along the wind and uphill slope vector to generate the +6h (Orange) and +12h (Yellow) forecast polygons, and computes a Hindcast Validation Score (%) by comparing predicted growth against historical satellite timestamps.

3. Complete Video Demo Script (Focus on ML & APIs)

Total Time: ~3 Minutes. Open http://127.0.0.1:8000/ before starting your recording.

Scene 1: Problem Statement & Architecture Intro (0:00 - 0:35)

[Screen Action: Show the main WildfireWatch map on Global (World) view.]
Voiceover:
"Hello everyone, this is WildfireWatch - our solution for Near-Real-Time Disaster Perimeter and Spread Forecasting. During a wildfire, satellites detect raw thermal hotspots, but emergency responders face two major challenges: first, raw satellite dots do not show the exact fire perimeter; and second, they do not predict where the fire will spread over the next 6 to 12 hours. WildfireWatch solves this by combining 4 live geospatial APIs with a 3-stage Machine Learning and Physics forecasting pipeline."

Scene 2: Live NASA FIRMS API & DBSCAN Clustering (0:35 - 1:20)

[Screen Action: Point cursor at Top Bar (Incidents, Hotspots, Latest Observation Date), then switch between 'Clustered Incidents' and 'Raw Hotspots' tabs on the left, and show the Region / Time Window dropdowns.]
Voiceover:
"Let's look at how data flows into our system. First, our FastAPI backend connects live to the NASA FIRMS REST API, ingesting real-time thermal anomalies from VIIRS and MODIS satellites across the globe. In the top bar, you can see the live incident count, total raw satellite hotspots, and the exact UTC timestamp of the latest satellite pass. You can filter the feed across 24, 48, or 72 hours, or focus on specific regions like India, North America, or Europe.
Instead of just plotting raw dots, our first Machine Learning stage runs Unsupervised DBSCAN Spatial Clustering using Haversine geodesic distance with a 15-kilometer epsilon radius. This automatically groups scattered satellite detections into distinct Wildfire Incidents and extracts the initial T+0h fire perimeter in hectares."

Scene 3: U-Net SWIR Verification, Weather/DEM APIs & Random Forest Forecast (1:20 - 2:30)

[Screen Action: Click on the top Wildfire Incident in the left sidebar. Let the map zoom into the Red (T+0h), Orange (+6h), and Yellow (+12h) polygons. Point cursor at the Forecast Details drawer on the left.]
Voiceover:
"When we click on any wildfire incident, our backend triggers the full multi-modal ML and API pipeline in real time:
First, for perimeter verification, we query the Element84 Sentinel-2 L2A STAC API and apply our U-Net Deep Learning segmentation pipeline trained via transfer learning on Short-Wave Infrared bands B12, B11, and B8A. Because Short-Wave Infrared penetrates thick smoke, the U-Net and Active Fire Index isolate the true burning perimeter - shown here as the solid red T+0h polygon.
Second, our backend makes live calls to the Open-Meteo Weather API and Digital Elevation Model API at the fire's exact coordinates to fetch real-time Wind Speed, Wind Direction, Relative Humidity, Air Temperature, and 5-point Terrain Slope.
Third, these live features along with NASA's Fire Radiative Power are fed into our 100-Tree Random Forest Spread Ensemble. The Random Forest predicts the forward Rate of Spread in meters per minute and couples with Rothermel's Elliptical Wavelet propagation model to generate the orange +6 hour forecast polygon and the yellow +12 hour forecast polygon."

Scene 4: Confidence Scores, Hindcast Validation & GeoJSON Export (2:30 - 3:00)

[Screen Action: Highlight Forecast Confidence, Hindcast Validation Score, Environmental Inputs, switch map layer top-right to Satellite view, and click 'Export GeoJSON'.]
Voiceover:

"In the details panel, we also provide two critical validation metrics: Forecast Confidence - computed from the variance across our 100 Random Forest trees and satellite sensor confidence - and a Hindcast Validation Score that measures spatial agreement between earlier and latest satellite passes. Finally, responders can switch to Satellite or Topographic map layers and click 'Export GeoJSON' to download standard RFC 7946 polygons for immediate use in GIS and emergency command systems. Thank you!"